FIELD EFFECT: MOVEMENT, SOUND AND SEMBLANCE IN AUDIO/VISUAL COMPOSITION

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ABSTRACT

This article takes Brian Massumi's examination of semblance and event as the starting point for building an audio/visual composition framework. *Field Effect,* a specific work arising from this investigation, is discussed and reflections on the composition process are offered. In particular, the problem of mapping from movement to sound in interactive works is considered, and strategies for examining the mapping process through content reduction and the use of non-local linkages are presented.

1. INTRODUCTION

This paper examines the process of creating an audio/visual installation based on human movement, using Brian Massumi's concepts of semblance and event.

Central to Massumi's philosophical framework is Benjamin's concept of *non-sensuous similarity* – the semblance of perception, which is to say, the experience of perceiving without having seen or heard. One example is the experience of having "seen" the extended trajectory of a fleeing mouse, even though you could not have actually seen where the mouse went (Massumi 2011, p. 17). A more sophisticated example draws upon Michel Chion's concept of *audiovision* in film, where, for example, a punch is thrown, and a victim staggers, and yet the film does not show the punch connecting, but only includes the sound of the hit. The audience will claim they saw the punch connect, even though it was never shown – the sound effect is experienced as a semblance of vision (Chion 1994).

Exploring this concept of non-local linkage, Massumi examines the way in which activation contours, or vitality affects, principally experienced in movement, detach themselves from their originating medium, and are experienced in other media, other forms (Massumi 2011, 106-108). The transfer of movement from one billiard ball to another has the same quality as a rear-end car collision, and the same quality can be perceived purely in sound, without the visual component. Despite the medium, the pattern, the rhythm of the movement is what remains.

In postulating the independence of such vitality affects, there are similarities with Scrimshaw's writing on the independence of affect in general (Scrimshaw 2013), and, drawing upon the rhythmanalysis of Lefevbre, a deep connection with sound itself (Lefevbre 2004).

All of which may appear to be purely speculative if not for the neuro-musicological evidence behind embodied sound cognition, which demonstrates the deep linkages that exist between movement, visual perception, and audition (Leman 2008).

Within the embodied sound cognition framework, there is an increasing body of work examining movement, gesture and sound (Godøy 2010; Altavilla et. Al. 2013) which in some respects can be understood as an investigation of the harmony between sound and movement.

It is important at this point to clarify that such a harmony is not based upon representational parallels between symbolic gestures, either in movement or sound, but at the deeper level of a shared activation contour, or more abstractly, a shared affective tonality. A rain dance does not look like or sound like rain, in any literal sense, and only over time does it become symbolically representational of rain – it originates in expressing the body's feeling of rain.

Creating new work, in whatever modality, frequently revolves around breaking through the accumulated symbolism in order to re-engage with the immediacy of direct experience. For example, Merce Cunningham's approach to choreography strips dance of "all representative and emotional elements that might drive movement… to focus on pure movement" (Gill 2011, p. 121).

"The dance is the conversion of the body's movement in space into the Time of its alteration: its speculative translation into a universe of pure bodily becoming. The life of the body unlimited, in a pure experience of its

becoming. Semblance of life. Embodied life: pure expression of the body's aliveness. Animatedness as such." (Massumi 2011, p. 141).

Taking this concept of "absolute movement" into music, Massumi suggests that absolute music is "music that fusionally mutually includes in an as its own pure event of expression all factors entering into its relational field, so that upon entering that field they *become* musical forces" (Massumi 2011, p. 145).

The concepts of "absolute movement" and "absolute music" are not to be understood as modes of expression which exist in isolation from all other forms. Conversely, they are are deeply relational – fusional – to the point that they "fold in" all other forms *in the moment of their own unfolding*, rather than rely upon prior knowledge, symbols or supporting text. In this sense, the impulse to create cross-modal works which share a common affective tonality is in itself the quest to express that which can only be expressed in the intersection of those modes, much in the same way that Chion's audiovisual punch does not exist without the specific interplay of vision and sound.

My own work within this framework of cross-modal expression encompasses gestural synthesis and interactive installation as well as visual-music composition (Pedersen 2012, 2013). The challenge in this cross-modal work frequently lies is exposing enough of the relationship between the different forms that a sense of the fusional nature of the work is apparent, while at the same time concealing enough of this same relationship that the work retains interest.

This is precisely the *semblance* at the heart of Massumi's investigation of Benjamin's concept of non-sensuous similarity, where the notion of concealing and revealing truth is central to the process of art making. Benjamin acknowledges the traditional link between beauty and truth, commenting that "what makes these truth-revealing sensuous forms evocative is their beauty, defined in a specific way: as a *harmony* of the sensuous forms involved" (Benjamin 1996, p. 224).

I have previously explored Benjamin's aesthetics with respect to the concepts of beauty and the sublime, and how this may apply to sound design, and found that an element of noise or dissonance is important in opening up a space in which to experience the sublime, as opposed to attempting to represent the sublime through perfection (Pedersen 2013). Likewise, Massumi is quick to distinguish between semblance which is symbolic (gesturing to some abstract truth), and semblance which is "quivering with life", "pure activity of life, abstractly appearing with all the reality of an intensive event," (Massumi 2011, p. 178).

What is most interesting in the crux of Massumi's argument is that these two aspects of semblance are polarities on a spectrum, not discrete dichotomies. Thus harmony still has a role to play. "Quivering life is never symbolic, but it can be forcibly contained in harmony and its symbolic associations. When it is, the harmony trembles with life, but does not release it… The expression of the truth hesitates, unable to complete the evocative retreat into the depths of reference. The *a priori* principle of resemblance waivers. Harmony is no longer guaranteed and authorized by it… Symbolism and metaphor are composed-away… Life quivers all the more intensely perceptually-felt, in a compelling openness of indetermination regained." (Massumi 2011, p. 178 – 179).

In applying Massumi's concepts of affective tonality, and in particular, the concepts of non-local linkage, field effect, semblance, harmony and openness to a new work, I've sought to develop an approach which addresses some of the challenges inherent in creating mappings between movement and sound, seeking the polarity of semblance that is still "quivering with life".

2. SOUND, MOVEMENT AND AFFECTIVE TONALITY

2.1. The composition framework

The works developed in this investigation centre around video recordings of movement by a dancer and the creation of an audio/visual composition derived from this recording[1].

In adopting framework for creating interactive sound and movement works through Massumi's concept of affective tonality, I've sought to use the following principles:

- Enable the activation contour jumping across media in order to create a shared affective tonality
- Employ indirect mapping which embodies the concept of a field effect, where many elements contribute to a whole
- Employ independent audio processes aimed at expressing only their own "occurence" yet containing the trace of external movement

The composition framework consists of three parts:

- A video analysis stage
- An intermediate "field effect" stage
- A surround sound rendering stage

The video analysis stage breaks the original recording into a 3x3 grid and extracts motion centroid and overall quantity of motion data from each cell of the grid.

[1] I gratefully acknowledge Ashlee Bye for permission to use a recording of her original choreography and performance, and the support of AusDance and the Newport Substation in providing the opportunity and the space in which to conduct this work.

The use of video analysis is not new in this kind of sound work, but personally the approach differs from earlier work I have done using the Kinect depth-sensing camera, where the movement analysis and sound synthesis made use of more direct, and also more symbolic, linkages (e.g. left-left hand position controls spectral content, right hand position controls spatialisation). Stepping away from tracking and labeling specific joints and specific gestures to focus purely on the quality and quantity of motion within the frame aligns the technical approach more strongly with the concept of bare activity underpinning the philosophical approach.

The intermediate field effect stage consists of a simulation of a 10 x 10 grid of linked particles which are anchored in a gravity-free 3D world by two central points, but otherwise are free to move. Individual particles are affected by forces generated by kinetic bodies moving through the plane of the grid, guided by the motion parameters extracted from the video analysis stage.

Individual particles collide with the kinetic bodies as they move, and they also collide with each other. The linkage constraints between the particles cause ripples to flow through the grid as a result of the original forces.

The audio rendering stage consists of an ambisonic encoder and decoder, with a 10x10 grid of source points in a horizontal plane. The target speaker system for rendering the work can be any number of speakers, but ideally uses no less than 16 channels in equidistant configuration around the circumference of the listening space. In this work, height information is not directly used, but overhead channels (even if irregular with respect to the horizontal configuration) have, in my experience, proven to aid the sense of immersion experienced by the listeners. A schematic of the overall composition framework is shown in Figure 1.

2.2. The composition

Collision data generated by the intermediate field effect stage is used to drive simple impulse generators connected to the 10x10 ambisonic source grid. This produces a spatialised field of clicks surrounding the listener, acting as an audible trace of the bare activity of the dancer in the original work.

Visualisation of the intermediate field effect stage forms the visual element of the composition. Where a multi-channel visualization is possible, a motiongram of the original movement is also presented.

This sparse approach to the sonic content of the work is a deliberate strategy of "composing away" any material which may be interpreted as symbolic or in any way extraneous to the bare activity of the dancer. In relation to "composing away", Massumi references the visual artwork of Robert Irwin, who continually stripped away elements of his work in order to find a point where the work became a "non-sensuous intensity of effect" (Massumi 2011, 158-159).

By reducing the sonic content to a mere impulse (a click) there is no consideration of timbre, pitch or envelope. This reduction assists in the investigation of the work as a fusion of movement and sound, in as much as the sound stimulus now only "makes sense" when experienced as audible spatialisation presented in conjunction with the visual trace of movement in field effect stage.

Just as Irwin gradually added additional elements back in to his visual work, timbre, pitch and duration elements can be added once the sense of a common affective tonality – a sense of fusion – has emerged.

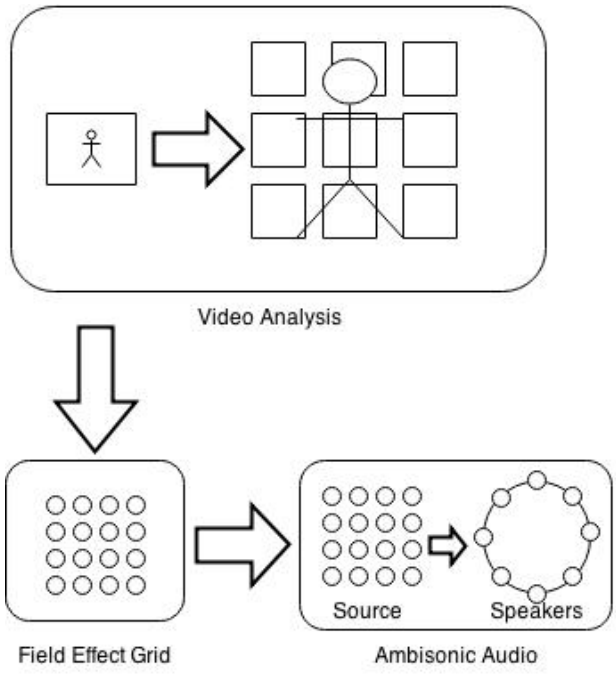


Figure 1. Schematic of the composition framework.

3. REFLECTIONS

In reflecting upon the compositional process, two elements stand out:

- focusing, if that is the right word, on the concept of the non-local linkage, or in other words, the indirect mapping
- reduction of content in order to foreground process

In my previous work with gesturally controlled synthesis, the vast array of possible mappings into complex synthesis space made the selection of only a few parameters seem like an impoverished option on the one hand, while the complexity of developing a more nuanced mapping that sonically reflected the intention

of the gesturer was overwhelming. It was difficult to find a productive middle ground.

By stripping away the majority of the available sonic parameters, I was able to focus more clearly on the raw quality of the original movement as it was traced in the audio layer. Reducing the dimensionality of the problem allowed me to evaluate the degree of semblance, of non-sensuous similarity, between the two sensuous modes.

One notable aspect of this reduced dimensionality was the way in which both the granularity of the visual analysis and the degree of spatialisation became important to maintaining the sense of linkage between the visual and audio modes. If the visual analysis is reduced to only a single centroid of motion, and the audio reduced to a mono channel, then the linkage was very weak. The perception is almost one of just a random clicking sound juxtaposed with a random jostling of the field effect grid.

As the granularity in both these dimensions increased, the sense of linkage increased[1]. With more nuanced analysis of the original video, movement qualities in different parts of the visual field translated across into multiple sites of activation within the field effect stage, and consequently as polyphony in the audio stage which matched the intensity envelope of the original choreography. Moving from mono to stereo to ambisonic surround sound, the sense of a shared affective tonality increased. In particular, my subjective experience of the moments of intensity in the choreography began to be matched by my experience of the immersive polyphony of clicks, even without watching the original movement recording. At this point the shared affective tonality that was missing with only a single motion centroid and a single channel of audio, began to emerge.

3.1. Further Work

The work done on this piece is only the beginning of a series of investigations into affective tonality in audio/visual works. In particular, examining how simple additions to the sound synthesis stage impact the composition process and the audience experience is an immediate next step. Using simple processes of subtractive synthesis (filtered white noise, with amplitude envelopes) will begin to slowly introduce elements of timbre, pitch and duration.

Alternate motion capture methodologies are also planned, including muscle activation sensors, inertial motion sensors, and Kinect point-cloud data, in order to see how the fidelity of the motion capture process impacts the overall composition framework.

Finally, further experimentation with the intermediate field effect stage is needed in order to see how an increase in dimensionality (e.g. a 3D cube of connected particles, rather than just a 2D grid) affects the overall process, particularly as the dimensionality of the synthesis space increases.

In addition to these technical aspects, the philosophical aspects of affective tonality and non-sensuous similarity warrant further consideration in the context of audio/visual art work, particularly where, as Robert Irwin does with his visual installation work, attention can be drawn no just to what is being perceived, but also to the process of perceiving. It is in this space that the harmony of our bodies and the sensuous forms of the work can lead to experiences of "liveness" which transcend any individual element, and from which new world-lines upon which to travel can emerge.

[1] This is my own purely subjective experience and not based on any kind of listener study.

4. REFERENCES

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